

Emran Yasser Moustafa

Experience

University of Edinburgh, PhD Student (Sept 2025 – onwards)

I have accepted a PhD position with the School of Informatics at the University of Edinburgh, supervised by Prof. Subramanian Ramamoorthy. My thesis is focused on modelling the behavior of vulnerable road users (pedestrians, cyclists, etc.) in dense urban environments. We intend on exploring how this information can be used to improve trajectory prediction and perform safer motion planning in mixed traffic.

Trinity College Dublin, Research Intern (June 2025 - Aug 2025)

I worked on model-based reinforcement learning under the supervision of Prof. Ivana Dusparic. First, I further developed my MEng thesis into a submission for IEEE ICAR 2025. I then worked on traffic signal control projects, specifically implementing and benchmarking a Mamba-based MBRL algorithm, Drama, against alternatives approaches in SUMO.

TU Delft, Lab Assistant (April 2024 – June 2024)

In this role, I worked with the new Swarming Lab in TU Delft. I collaborated with the lab staff to improve their ROS-based swarming stack. I also collaborated with the Science Center in TU Delft to develop a swarming exhibit for their “travelling science gallery” project, titled Science on Tour.

Trinity College Dublin and Flying Labs Madrid, Research Intern (June 2023 - Sept 2023)

I worked two six week internships in TCD and Flying Labs Madrid (affiliated w/ UC3M) under the supervision of Dr. Harun Siljak and Dr. Jesus Garcia, respectively. In Dublin, I developed an open-source drone swarm control package for using ROS. The package is currently being used in the curriculum for Master’s engineering modules in TCD. In Madrid, I prototyped and benchmarked a novel visual-inertial SLAM system. My experiments focused on deployment on Crazyflie drones. The system used SuperPoint and SuperGlue, and was built with C++, OpenGL and ROS.

Trinity College Dublin, Research Intern (July 2022 - Sept 2022)

My research, supervised by Dr. Shreejith Shanker, was centred around the use of FPGA and reconfigurable computers for object detection in autonomous vehicles. I developed a computer vision model for an FSAI vehicle and deployed it on a Xilinx KV260. I then benchmarked its performance against other more widely used compute platforms. I used Pytorch and OpenCV for the testing and development of my model.

Education

Trinity College Dublin, Computer and Electronic Engineering BEng. & MEng. 2025

1st Year Grade: 80%, First Class Honours

2nd Year Grade: 73%, First Class Honours

3rd Year Grade: 79%, First Class Honours

5th Year Grade: 83%, First Class Honours

TU Delft, Robotics MSc. 2024, Erasmus Exchange

4th Year Grade: 8.7/10.0, First Class Honours

Contact

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Soft Skills

- Project Management
- Product Management
- Leadership
- Fundraising
- Grant Writing

Technical Skills

- Python / C / C++
- ARM Assembly
- Verilog
- MATLAB
- Robot Operating System
- OpenCV
- Pytorch / TensorFlow
- MMDetection3D
- Docker
- Engineering Research

Areas of Expertise

- Reinforcement Learning
- Planning & Control
- 2D/3D Computer Vision
- Localisation & Mapping
- Image Processing
- Deep Learning / Machine Learning
- ADAS Software

Volunteering

Do Sheal, Co-Founder (June 2025 – onwards)

Do Sheal is an indie board game studio based in Dublin, Ireland. With a team is comprised of artists and scientists, our goal is to create board games based on strategic dynamics that develop from real-world phenomena. Funded via a grant from the EU, we are currently designing a climate-crisis themed board game titled, titled Twelve Seconds to Midnight. I am involved in the project primarily in a project management capacity.

Formula Trinity, AI Team Captain (July 2022 – July 2023)

I managed a team of 30 students to develop an autonomous driving system for Formula Student UK 2023. My role on the team was a combination of project management and product development. Overall, we placed 8th out of 20 teams from around Europe and the UK. Prior to this (Oct 2021-July 2022) I lead the state estimation team, where we built a FastSLAM1 package for the vehicle as well as other tools for localization.

RacingAI, Founder (May 2022 – Sept 2023)

I founded a non-profit organisation called RacingAI with the mission of growing the robotics and autonomous navigation industries here in Ireland. We hosted events, workshops and competitions as well as worked with student organisations to help them grow the robotics community in their own colleges. In our first year, I raised a total of €17,000 to run two projects with the organization.

Awards

University of Edinburgh, UKRI Scholarship (£90,000 over three years)

- Awarded to fund my PhD with the School of Informatics, including tuition and an annual stipend

Trinity College Dublin, Collen Prize for Computer Engineering

- Awarded for the best thesis project in Computer Engineering

Trinity College Dublin, Gold Medal for Electronic and Computer Engineering

- Awarded for achieving >80% in my final grade for the BEng Electronic and Computer Engineering

Jones Engineering, Eric and Barbara Kinsella Scholarship 2021 (€20,000 over four years)

- Awarded on academic and leadership merit

Laidlaw Foundation, Leadership and Research Scholarship 2022 (€10,000 over two years)

- Awarded on academic and leadership merit

Trinity College Dublin, Dean of Students' Leadership Award for Volunteering

- Organised and hosted intervarsity F1Tenth competition in the University of Edinburgh (EUFS vs Formula Trinity)

Trinity College Dublin, Stanford-Smith Prize for Electronic and Computer Engineering

- Highest overall grades in Electronic and Computer Engineering, 2022/23

Trinity College Dublin, ER. Stuart Prize for Engineering

- Highest overall grade in Chemistry for Engineering course

Projects

Model-Based Reinforcement Learning for Multi-Agent Autonomous Racing, Trinity College Dublin

For my master's thesis, I explored the use of MBRL algorithms in the domain of adversarial multi-agent autonomous racing. I parameterised the behaviour (speed & steering) of adversarial agents in the F1Tenth environment with an episode context. I then trained several MBRL algorithms in this environment, including a novel context-aware method. Using the episode's context, one could then induce out-of-distribution dynamics in the environment and infer each algorithm's capability to generalise and adapt to unseen driving behaviours. My proposed algorithm demonstrated improved performance in agent-to-agent interactions, as evidenced by an increased rate of overtakes and reduced rate of agent-to-agent collisions. This work culminated in a paper submitted to **IEEE International Conference of Advanced Robotics**, where I featured as the first author.

3D Object Tracking: From Front to Surround View, TU Delft

Worked with the Intelligent Vehicles group at TU Delft on a 15 ECTS research assignment. I was challenged with developing a 3D auto labelling pipeline for the View-of-Delft dataset, an autonomous driving dataset recorded in the historic city of Delft. My system detected and tracked multiple road users from an input point cloud sequence, extending the existing trajectories annotations from the front view of the vehicle to 360 degrees around the vehicle. My project received a grade of **9.0/10.0**. My work was also included in a paper the group submitted to **IEEE Robotics and Automation Letters**, where I featured as the second author.

FS-AI UK 2022 and FS-AI UK 2023, Formula Trinity

I competed in FS-AI UK in both 2022 and 2023. We came **5th** and **8th** in those two years. During my stay on the team, I touched all aspects of both the design and the organisation. I worked both as an engineer and in management. In those two years, we won both Newcomer of the Year and the Real-World AI award. I was also nominated by my team for the Craig Dawson Award for most valuable team member.

ROS FastSLAM1 Module, Formula Trinity

With the help of two other students on the FT state estimation team, we developed a FastSLAM1 module in ROS2. I coordinated the design in collaboration with other departments. Our algorithm was written in Python and evaluated in simulation using RViz as well as on a physical testing rig. Come competition time, the design was successfully deployed on the FS-AI vehicle and had a latency of **44 FPS**.

AI Grand Prix and F1Tenth, Formula Trinity

Early in my robotics journey, I developed an autonomous racing algorithm for the F1Tenth-based AI Grand Prix competition. This was an online autonomous racing competition hosted by the newly formed Formula Trinity. My solution was an adaptation of the disparity extender algorithm. Myself and my team placed 3rd in the competition, out of 16 teams, winning a prize of **€200**. A few months later, Formula Trinity and I then used a similar algorithm in the F1Tenth competition at **IROS 2021**, Prague, Czech Republic.

Hardware Accelerated Perception Module for FS-AI Vehicle, Trinity College Dublin

Working in Trinity's Electronic and Electrical Department, I investigated the use of FPGAs for object detection in high-speed autonomous vehicles. In practice I created a PyTorch computer vision model for detecting FS-AI cones and deployed it on Xilinx's Kria SoM. I developed and trained a segmentation model, based primarily on CGNet. The model had approximately 600K parameters, with a speed of **15 FPS** and **0.72 IoU**.

rclpy_crazyflie, Trinity College Dublin

rclpy_crazyflie is a ROS2-wrapper for the Crazyflie API. I developed this product with the primary goal of creating an easy-to-use platform for MAI students in Trinity College to communicate and control the Crazyflie drones using ROS. The primary benefit of this package is its compatibility with other ROS and robotics systems, as well as the drone swarm control capabilities it offers users. As far as I am currently aware (10/2024), this package is being used by Master's students in both Trinity College Dublin and Lund University, Sweden.

References

Provided on request. More information can be found on my LinkedIn.